

CLASS XI · LECTURE 4

Software and Language Translators

One to two periods · Compiler and interpreter is a standard OR question

By the end of this lecture you can

- Classify software as system software, application software or a utility.
- Place a language as machine, assembly or high-level.
- Explain why a compiler is more efficient at run time than an interpreter.
- Define source code and object code.

Three families of software

System software runs the machine and gives other programs a place to stand. Application software does a job the user came for. A utility is a house-keeping tool. If you sort a program into the wrong family, the rest of the answer drifts. Learn the families before the brand names.

Family	Purpose	Examples
System software	Controls hardware and offers services to other programs	Operating system, device drivers, language translators
Application software	Does a user's task	A browser, a payroll program, a game, a word processor
Utility	Maintains the system	Antivirus, compression, disk checker, backup tool

Application software is sometimes split into **general purpose** (a spreadsheet, used by many kinds of people) and **special purpose** (a program written for one organisation, such as a college's own fee system). Device drivers are system software: they teach the operating system how to talk to one piece of hardware, such as a particular printer. Antivirus is a utility, and a virus itself is software, not hardware. That one-mark fact is easy to lose.

Generations of programming language

A program is written in a language. The processor only understands machine language: numbers that mean operations and addresses. Everything else must be translated.

Language	Looks like	Who translates it
Machine language	Binary instructions for one family of CPU	Nothing. The CPU runs it
Assembly language	Short symbols such as ADD and MOV, one per machine instruction	An assembler
High-level language	Statements nearer to English or algebra, such as C, Visual Basic or C++	A compiler or an interpreter

DEFINITION Source code and object code

Source code is the program as the programmer wrote it. Object code is the machine-language result after translation. A linker may join object code with library code to make the executable program.

Compiler and interpreter

Both turn a high-level program into something the machine can run. They do it at different times, and that is the whole of the comparison the paper wants.

Point	Compiler	Interpreter
When it translates	The whole program, before it runs	One statement at a time, while it runs
What you keep	Object code, which can be run again without translating	Usually no separate saved object program
Errors	A list of errors after compilation	Stops at the first error it meets
Speed of the run	Faster, because translation is already finished	Slower, because translation happens on every run
Helpful for	A finished program you will run many times	Testing and learning, line by line

ON THE BOARD Why a compiler is more efficient

A compiler translates the source program once and saves object code. Each later run executes that object code directly, so the user does not pay the translation cost again. An interpreter translates during every run, statement by statement, so the same program takes longer. That is why a compiler is more efficient at execution time.

An **assembler** is the translator for assembly language, not for C or Visual Basic. Do not call a compiler an assembler. C in Paper II is a compiled language. Visual Basic in the college lab is usually run through a compiler as well, though the environment also lets you test as you type.

EXAM TIP Do not swap the words

Machine language needs no translator. Assembly language needs an assembler. A high-level language needs a compiler or an interpreter. "Compiler" is not a general word for every translator.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. A computer virus is:

- A) Hardware
- B) Software
- C) A port
- D) A bus

2. Assembly language is translated by:

- A) A compiler
- B) An assembler
- C) A plotter
- D) A browser

Short answers

S1. How is a compiler more efficient than an interpreter?

S2. Differentiate system software and application software, with one example of each.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. A virus is a program. Harmful, but still software.

MCQ 2. B. Compiler and interpreter are for high-level languages.

S1. A compiler translates the whole source program into object code before execution, and that object code can be run again without translating. An interpreter translates one statement at a time during every run. The compiled program therefore executes faster, so the compiler is more efficient at run time.

S2. System software controls the hardware and provides services for other programs. The operating system is system software. Application software does a task the user wants, such as a word processor or a college fee program. The computer needs system software before an application can run.